PROJECT SYNOPSIS

ON

LostLink: An Intelligent Lost-and-Found System with Multi-Modal Matching and Evidence-Based Ownership Verification

Submitted to the Department of Computer Science and Engineering
in Partial Fulfillment of the Requirements
for the Degree of

Bachelor of Technology

in

Computer Science and Engineering

by

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JSS ACADEMY OF TECHNICAL EDUCATION, NOIDA
DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADESH, LUCKNOW
(Formerly Uttar Pradesh Technical University, Lucknow)

SEPTEMBER, 2026

Vision and Mission

Vision of the Institute

JSS Academy of Technical Education Noida aims to become an Institution of excellence in imparting quality Outcome Based Education that empowers the young generation with Knowledge, Skills, Research, Aptitude and Ethical values to solve Contemporary Challenging Problems.

Mission of the Institute

1. Develop a platform for achieving globally acceptable level of intellectual acumen and technological competence.
2. Create an inspiring ambience that raises the motivation level for conducting quality research.
3. Provide an environment for acquiring ethical values and positive attitude.

Vision of the Department

"To spark the imagination of the Computer Science Engineers with values, skills and creativity to solve the real-world problems."

Mission of the Department

1. To inculcate creative thinking and problem-solving skills through effective teaching, learning and research.
2. To empower professionals with core competency in the field of Computer Science and Engineering.
3. To foster independent and lifelong learning with ethical and social responsibilities.

Program Outcomes (POs)

Engineering Graduates will be able to:

PO1: Engineering knowledge

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and teamwork

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning

Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Educational Outcomes (PEOs)

PEO1 To apply computational skills necessary to analyze, formulate and solve engineering problems.

PEO2 To establish as entrepreneurs, and work in interdisciplinary research and development organizations as an individual or in a team.

PEO3 To inculcate ethical values and leadership qualities in students to have a successful career.

PEO4 To develop analytical thinking that helps them to comprehend and solve real-world problems and inherit the attitude of lifelong learning for pursuing higher education.

Program Specific Outcomes (PSOs)

PSO1 Acquiring in-depth knowledge of theoretical foundations and issues in Computer Science to induce learning abilities for developing computational skills.

PSO2 Ability to analyse, design, develop, test and manage complex software system and applications using advanced tools and techniques.

Course Outcomes and CO–PO–PSO Mapping

Course Outcomes (COs)

C410.1 Identify, formulate, design and analyze a research based/web based problem.

C410.2 Communicate effectively in verbal and written form.

C410.3 Apply appropriate computing and engineering skills for obtaining solution to the formulated problem within a stipulated time.

C410.4 Work effectively as a part of team in multi-disciplinary areas.

C410.5 Consolidate the final outcome in the form of a publication.

CO–PO–PSO Mapping

Table 1: CO–PO–PSO mapping matrix (correlation levels 1–3)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C410.1	3	3	3	3	2	3	3	3	3	3	2	3	3	3
C410.2	2	2	2	2	2	2	2	2	2	3	2	3	2	3
C410.3	3	3	3	3	3	3	2	3	3	3	3	3	3	3
C410.4	3	3	3	3	2	3	2	3	3	3	3	3	3	3
C410.5	3	3	3	3	3	3	3	3	3	3	3	3	3	3
C410	2.80	2.80	2.80	2.80	2.40	2.80	2.40	2.80	2.80	3.00	2.60	3.00	2.80	3.00

Declaration

We hereby declare that this synopsis submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Certificate

This is to certify that the project synopsis report entitled **"LostLink: An Intelligent Lost-and-Found System with Multi-Modal Matching and Evidence-Based Ownership Verification"** which is submitted by Rohit Kumar, Yuvraj Mishra and Aman Saleem in partial fulfillment of the requirement for the award of degree B.Tech. in the Department of Computer Science and Engineering of Dr. APJ Abdul Kalam Technical University, Uttar Pradesh, Lucknow is a record of the candidates' own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

_____

Signature

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Date: _____

Abstract

Lost-and-found recovery on campuses today relies almost entirely on notice boards, informal messaging groups, and keyword-driven listing portals, which struggle to surface the single genuine match among many similar-looking reports and offer no reliable way to confirm that a claimant is the true owner of a recovered item. **LostLink** is proposed as a web-based platform that addresses both halves of this problem. On the matching side, it uses zero-shot, similarity-native visual embedding models (CLIP and DINOv2) rather than category-specific fine-tuned networks, so that image similarity can be evaluated across a broad range of item categories without collecting a fresh labelled dataset for each one; this visual signal is fused with text, location, and time similarity into a single weighted, explainable match score. On the verification side, LostLink introduces a scored, evidence-based ownership-verification mechanism in which a claimant must supply private identifying details not shown on the public listing, evaluated using the False Acceptance Rate / False Rejection Rate framework standard in authentication research, with multi-claimant disputes resolved by ranking. Matching performance will be evaluated using precision, recall, F1-score, and top-K retrieval accuracy against a keyword/category baseline; verification performance will be evaluated using FAR/FRR across thresholds and identification-rate-at-rank for contested claims. The project targets a centralised, transparent, and auditable recovery workflow, closing two gaps identified in the existing literature: the absence of a cross-category zero-shot matching evaluation, and the absence of a quantitatively measured ownership-verification mechanism for lost-and-found claims.

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List of Symbols and Abbreviations

Symbol/Abbreviation	Description
CLIP	Contrastive Language–Image Pre-training
DINOv2	Self-Distillation with No Labels, version 2
CNN	Convolutional Neural Network
FAR	False Acceptance Rate
FRR	False Rejection Rate
EER	Equal Error Rate
API	Application Programming Interface
JWT	JSON Web Token
TF-IDF	Term Frequency–Inverse Document Frequency
UI/UX	User Interface / User Experience
DB	Database
ML	Machine Learning
RQ	Research Question

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Chapter 1

Introduction

Personal belongings are lost every day in colleges, offices, and public places. Recovery today relies almost entirely on notice boards, social media groups, and manual word-of-mouth. Existing digital lost-and-found tools mostly provide a searchable list of lost and found reports, but finding the right match among many similar-looking entries remains genuinely difficult, and confirming that a claimant is the true owner of a recovered item is even harder.

LostLink is proposed as an intelligent lost-and-found system that combines image, text, location, and time information to surface likely matches automatically, and adds a separate, evidence-based verification step to confirm that a claimant is the real owner rather than simply the first or most convincing person to respond. The system is intended to reduce the time and effort required to recover lost belongings while making the recovery process transparent and auditable for administrators.

1.1 Problem Statement

Several distinct difficulties make lost-and-found recovery unreliable in practice:

- Relevant lost or found reports are difficult to find among many listings.
- Keyword search misses items that are visually similar but described in different words.
- Two people frequently describe the same object very differently.
- Location and time context is collected by existing systems but rarely used in matching.
- Multiple people may claim the same found item, with no reliable way to compare their claims.
- Manually verifying legitimate ownership is slow, inconsistent, and depends entirely on administrator judgement.

This motivates the central question: *how can an intelligent system identify potential matches and verify legitimate ownership, while reducing false and fraudulent claims?*

1.2 Motivation

- Reduce the time required to recover lost belongings.
- Reduce dependence on manual searching across notice boards and social media groups.
- Improve matching accuracy by using multiple information sources together — image, text, location, and time — instead of keywords alone.
- Reduce fraudulent or incorrect ownership claims through a structured verification mechanism.
- Provide one centralized recovery workflow instead of scattered, informal channels.
- Keep the recovery process transparent and auditable, so administrators can review why a match or a claim was accepted.

Chapter 2

Objective(s)

2.1 Primary Objectives

- Develop a centralised lost-and-found platform.
- Automatically surface potential matches from new reports.
- Use pretrained computer-vision models for image similarity.
- Combine visual and contextual signals — text, location, and time.
- Build a scoring-based ownership-verification mechanism.
- Reduce false and fraudulent claims.

2.2 Research Objectives

RQ1 — How do zero-shot embeddings (CLIP, DINOv2) compare to trained, category-specific fusion models for cross-category lost-and-found matching?

RQ2 — Can a scored, evidence-based verification mechanism measurably lower the false-acceptance rate for fraudulent claims without a large rise in false rejections of genuine owners?

2.3 Proposed Pipeline

Figure 2.1 summarises the five-stage pipeline through which every report and claim passes.

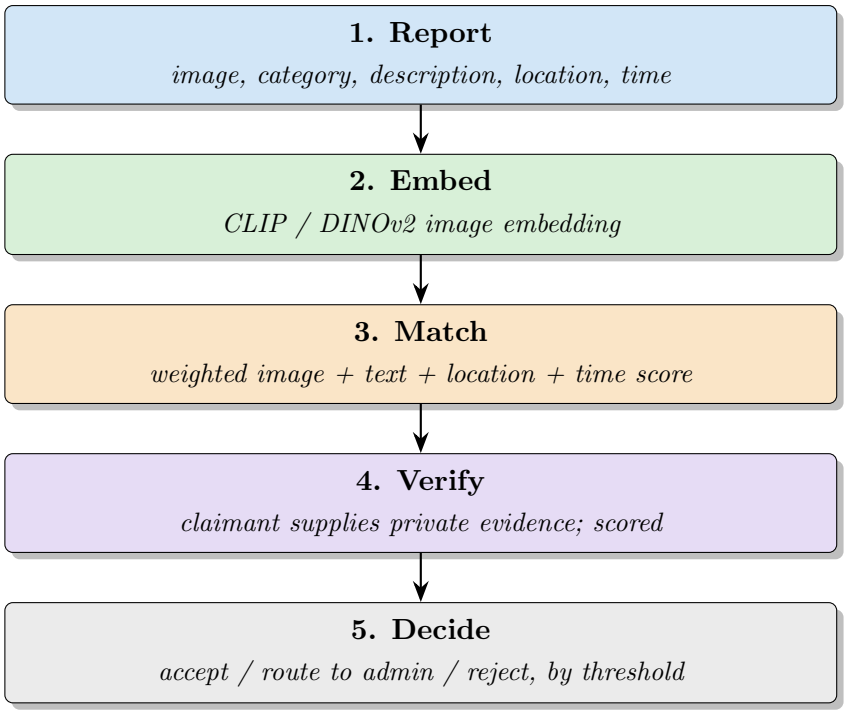


Figure 2.1: The five-stage LostLink pipeline, from report submission to a decision on a claim.

Chapter 3

Scope of the Project

3.1 In Scope

- Web application front end (React).
- Lost and found reporting with image upload.
- Image similarity using pretrained CLIP / DINOv2 embeddings.
- Text, location, and time similarity.
- Multi-factor match-score generation.
- Ownership verification and scoring.
- Multiple-claim ranking.
- Notifications for potential matches and claim status.
- Administrator dashboard for review and approval.
- Recovery history and audit trail.

3.2 Proposed System Modules

The system is organised into the following modules: user registration and authentication; lost-item reporting; found-item reporting; image embedding generation (CLIP / DINOv2); text similarity analysis; location and time matching; multi-factor match-score generation; potential-match notification; ownership verification; claim management and ranking; administrator verification dashboard; and secure handover and recovery history.

Chapter 4

Related Work

A review of current lost-and-found and image-matching literature shows that several relevant systems already exist, though each addresses only part of the problem LostLink is designed to solve.

Table 4.1: Comparison of existing lost-and-found and image-matching systems

System	What it does	Where it falls short
Dhanawardhana et al. (2025) [1]	MobileNetV2-based Siamese network, fine-tuned on 848 bag-image pairs; fuses image, text, location and time similarity into one score.	Fine-tuning ties the model to a single item category (bags); no ownership-verification component.
LostNet (Zhou et al., 2024) [2]	MobileNetV2 with CBAM attention, fine-tuned for image matching, combined with perceptual hashing.	Image-only; single-category focus; no text, location, time, or verification signal.
Campus claim-arbitration system (2026, preprint) [6]	Describes an ownership-verification and multi-claim arbitration workflow for campus lost-and-found.	Non-peer-reviewed; no quantified fraud-detection or verification-accuracy metrics published.
Typical campus lost-and-found portals [7, 8]	Registration, keyword/category search, notifications, and an administrator panel.	Matching is keyword or category based; the correct item is easy to miss among similar listings.

Keyword- and metadata-based matching. Most deployed lost-and-found portals, including typical campus systems, are built around registration, a keyword or category search form, and an administrator panel that manually reviews reports [8, 7]. UniFind layers natural-language processing and basic image recognition on top of this structure, but verifies claimants through college login credentials rather than through evidence tied to the item itself [7]. Map-based systems add location context to listings, though matching remains driven by manually entered fields [16], and blockchain-based proposals focus on a tamper-resistant audit trail while still leaving matching to keyword search [17]. The shared limitation is that two people rarely describe the same object in the same words, so keyword matching misses genuine matches whenever descriptions diverge.

Fine-tuned convolutional and Siamese matching. The Siamese-network idea, introduced by Koch et al. for one-shot image recognition [10], underlies most category-specific lost-and-found matchers. Dhanawardhana et al. fine-tune a MobileNetV2-based [13] Siamese network on 848 hand-labelled bag-image pairs and fuse the resulting score with

text, location, and time similarity [1]. LostNet combines a MobileNetV2 backbone with attention and perceptual hashing, reporting 96.8% test accuracy [2]. A separate ResNet-50-based similarity study reports only around 30% accuracy, illustrating that a classification-oriented backbone does not automatically transfer to open-set similarity matching [11]. The shared constraint is that a Siamese network's usefulness depends entirely on the category it was fine-tuned on — extending it to a new category requires a fresh labelled dataset.

Zero-shot vision-language and self-supervised embeddings. CLIP is trained contrastively on image–caption pairs so that related images and text land close together in a shared embedding space, performing competitively with no task-specific fine-tuning [3]. DINOv2 instead uses self-supervised distillation to produce general-purpose visual features without labels [4]. The two are not interchangeable: self-supervised encoders such as DINOv2 tend to separate visually similar instances of the same category more cleanly, while CLIP is comparatively stronger when textual cues drive the match [19, 18]. A CLIP-based institutional prototype (ViT-L/14) reports 95.2% accuracy matching images to text descriptions on a 500-image CIFAR-10 subset [9] — substantially stronger than the fine-tuned CNN baselines above, though on a generic benchmark rather than the near-duplicate, single-category confusions typical of real lost-and-found reports. No system reviewed here reports a head-to-head CLIP-versus-DINOv2 comparison on lost-and-found imagery specifically.

Multimodal fusion. Image similarity alone is insufficient, since two different black backpacks can look nearly identical while location and time context can disambiguate them. General strategies for combining image and text embeddings are well studied [5], and fusing features across multiple encoder layers, rather than only the final embedding, has been shown to improve cross-modal retrieval [12]. Within lost-and-found specifically, Dhanawardhana et al.'s weighted fusion of image, text, location, and time similarity is the clearest precedent [1], though the per-modality contribution is rarely isolated and measured.

Ownership verification. Once a candidate match is surfaced, most existing prototypes stop short of verifying the claimant is the true owner. UniFind verifies claimants through college-ID login rather than item-specific evidence [7]; a 2026 preprint describes a claim-arbitration workflow but reports no measured accuracy [6]; and a broader survey attributes a large share of never-recovered campus items to the absence of a reliable ownership check [8]. Outside this literature, verifying a claimed identity from supplied evidence is a well-studied authentication problem, evaluated using False Acceptance Rate (FAR) and False Rejection Rate (FRR) across thresholds [14], with benchmarks such as EVI applying this to knowledge-based verification and reporting identification-rate-at-rank for multi-claimant scenarios [15].

4.1 Conclusion

Two gaps stand out from this review. **First**, matching systems that report strong accuracy achieve it by fine-tuning on a single object category [1, 2]; whether zero-shot embeddings, and DINOv2 in particular, can match or exceed this accuracy across a broad range of item categories *without* any fine-tuning remains an open, testable question. **Second**, ownership verification in lost-and-found systems is currently either absent, reduced to a login check, or described only as a workflow without measured performance [7, 6], while a standard, threshold-based FAR/FRR evaluation framework already exists in the authentication literature and has simply not been applied here [14, 15]. LostLink is designed to close both gaps with measurable, cross-category results rather than a further descriptive prototype.

4.2 Future Scope

- Mobile application.
- QR-code-based item registration.
- Integration with campus security systems.
- A cross-campus recovery network.
- Fine-tuning of embedding models if a sufficiently large in-domain dataset becomes available.

Chapter 5

Hardware and Software Requirements

5.1 Software Requirements

Table 5.1: Technologies and tools used

Category	**Tools / Technologies**
Frontend	React.js, Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB
ML Service	Python
Image Processing	OpenCV
Image Embeddings	CLIP, DINOv2
Similarity Computation	Cosine similarity
Text Matching	TF-IDF / sentence embeddings
Maps	Leaflet + OpenStreetMap, or Google Maps API
Authentication	JWT
Notifications	Firebase Cloud Messaging, Email
Version Control / Deployment	Git, GitHub; Vercel (frontend); Render/Railway (backend); MongoDB Atlas

5.2 Hardware Requirements

- **Development machine:** a laptop/desktop with at least 8 GB RAM and a modern multi-core CPU is sufficient for frontend and backend development; a CUDA-capable GPU (e.g. 6 GB+ VRAM) is recommended, though not required, for faster local experimentation with CLIP/DINOv2 embedding generation.
- **Inference/deployment:** embedding generation can run on CPU for the expected report volume of a campus-scale deployment; a small GPU-backed instance (e.g. a free/low-tier cloud GPU) would reduce latency if report volume grows.
- **Storage:** standard cloud object storage or the deployment platform's file storage for uploaded item photographs; MongoDB Atlas free tier is sufficient for structured data during development.

- **Network:** a stable internet connection for API calls between the frontend, backend, and the Python ML microservice, and for third-party services (maps, notifications).

5.3 System Architecture

Figure 5.1 shows the layered architecture in which the client application communicates with a Node.js API, which delegates image and text similarity computation to a dedicated Python machine-learning service.

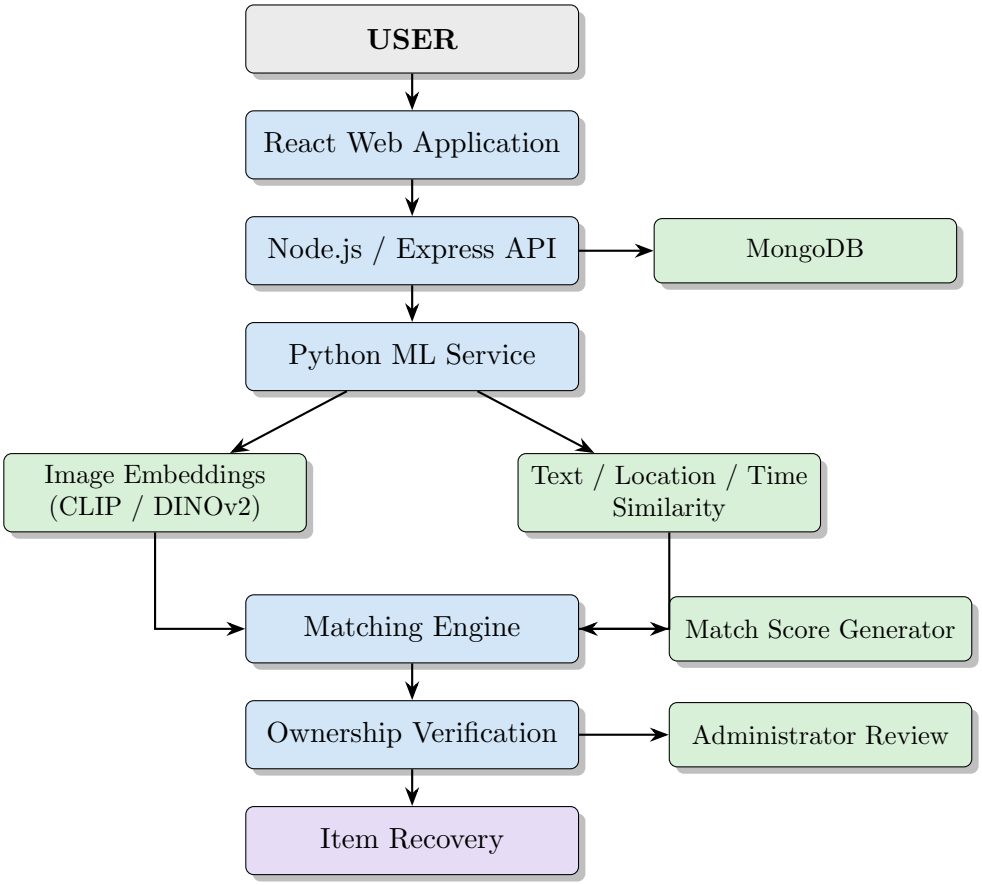


Figure 5.1: LostLink system architecture / block diagram.

The core matching computation combines four weighted similarity signals into a single score:

$$S = w_1 S_{\text{image}} + w_2 S_{\text{text}} + w_3 S_{\text{location}} + w_4 S_{\text{time}}, \qquad \sum_i w_i = 1 \tag{5.1}$$

with the weights w_i tuned on a held-out validation split rather than fixed a priori, following the fusion structure used by Dhanawardhana et al. [1].

Chapter 6

Predicted Timeline for the Project

The project is planned over a six-month window from August 2026 to January 2027, as shown in Figure 6.1.

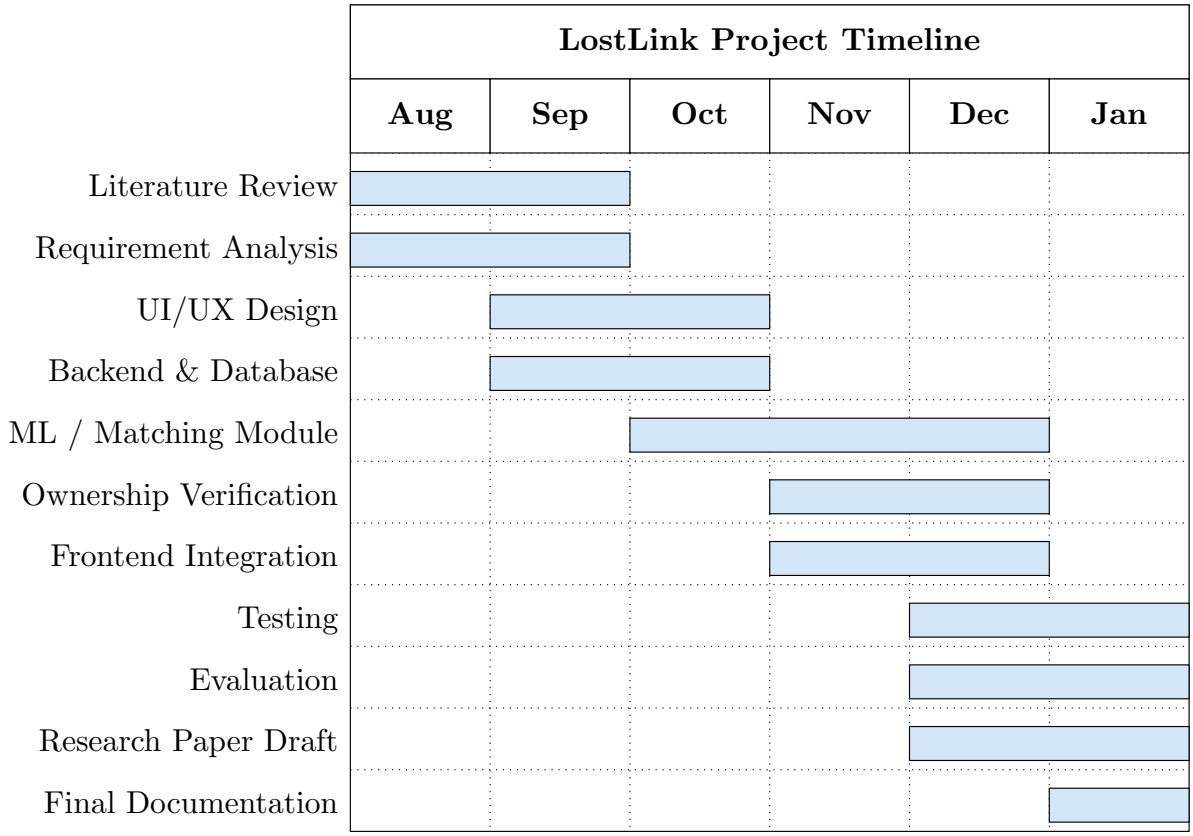


Figure 6.1: Project timeline (Gantt chart), August 2026 – January 2027.

Table 6.1: Project phases and their planned months

Phase	**Planned Months**
Literature Review	Aug – Sep
Requirement Analysis	Aug – Sep
UI/UX Design	Sep – Oct
Backend & Database	Sep – Oct
ML / Matching Module	Oct – Dec
Ownership Verification	Nov – Dec
Frontend Integration	Nov – Dec
Testing	Dec – Jan
Evaluation	Dec – Jan
Research Paper Draft	Dec – Jan
Final Documentation	Jan

Chapter 7

Conclusion

Lost-and-found recovery has been studied from several angles: keyword and metadata search, category-specific Siamese image matching, general-purpose zero-shot embeddings, multimodal fusion of image with text, location and time, and, least developed of all, ownership verification. Convolutional Siamese networks deliver strong accuracy but do not generalise past the category they were trained on, while zero-shot vision-language and self-supervised models promise to remove that constraint at the cost of being largely untested for this specific application. Ownership verification remains the least mature area: existing systems either skip it, substitute a login check, or describe it conceptually, without reporting the false-acceptance and false-rejection rates that authentication research treats as the standard way to evaluate exactly this kind of mechanism.

LostLink is designed to close both gaps. Matching performance will be evaluated using precision, recall, F1-score, and top-K retrieval accuracy, comparing CLIP, DINOv2, and a keyword/category baseline, both within a single item category and across categories. Ownership-verification performance will be evaluated using False Acceptance Rate and False Rejection Rate at varying thresholds, and multi-claimant ranking will be evaluated by identification-rate-at-rank. Based on accuracy figures gathered across the reviewed literature (roughly 75–97% depending on task and category breadth), none of which was measured on the specific, cross-category, near-duplicate matching problem LostLink addresses, the working target for the matching pipeline is 90–95% accuracy across multiple item categories, to be evaluated once the system is built and tested rather than assumed in advance.

Future work, once the system has been implemented and evaluated, includes extending the item-category set, testing fine-tuning of the embedding models if a sufficiently large in-domain dataset becomes available, and studying whether the explainable score breakdown shown to users changes how administrators resolve contested claims in practice.

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Appendix A

Plagiarism Report

A.1 Plagiarism Report

[To be inserted after the plagiarism-check tool provided by the institute has been run on the final report.]

A.2 AI Content Report

[To be inserted after the AI-content-detection tool provided by the institute has been run on the final report.]